Waste Right

Household waste collection, starting at ₹10

Scan your waste.
Book a pickup.
Know what happens next.



Zenith Hackathon / 24 October 2026

Problem statement

Everyday waste needs a simpler path from the home to the right processor.

How can a household identify mixed waste, arrange an affordable pickup and know what was actually collected?

Waste Right combines phone-camera assistance with a human-verified collection service.

Pain points for residents and helpers

The difficulty continues after someone identifies the waste.

For residents

Mixed packaging is confusing.
Collection takes too many steps.
Weight and final fees are unclear.
The outcome is hard to track.

For collection partners

Jobs and addresses are scattered.
Material and weight need checking.
Earnings and dues need clear records.
Disputes need a simple process.

The solutions we provide

Each feature removes a specific step or uncertainty.

Less effort to start

Scan with a phone or describe the waste. Confirm the suggestions.

A clear collection flow

Reuse an address, choose a time and see automatic charges. The helper records weight.

A shared record

Residents review receipts. Helpers see earnings. Operations handles payments and issues.

Complete Waste Right overview

One service connects residents, helpers, Operations and processing partners.

Resident	Scan → confirm items → schedule → accept the collection receipt.
Helper and Operations	Collect → weigh → verify → record earnings, payments and disputes.
Processing and benefits	Hand over accepted material → record actual recovery → apply earned credits and rewards.

AI suggests. People confirm. Every financial benefit needs a record.

The resident journey

A short flow, from the camera to the collection receipt.

01 Scan and confirm Use the phone camera or a description. Correct the suggested items.

02 Schedule Choose the address, date and time. Check the automatic fee.

03 Review the receipt The helper weighs the waste. Accept the receipt or report a problem.

The user keeps the final say

A camera makes starting easier. Confirmation makes the record trustworthy.

What the app does

Suggests items and materials.
Reads visible labels with OCR.
Asks for clarification when needed.

What the resident controls

Removes a mistaken item.
Corrects mixed materials.
Confirms what will be collected.

Photographs do not establish weight or safe handling.

One mixed-waste example

Illustrative journey, not a recorded customer collection.

Camera view	Fruit scraps and a plastic food container appear together.
Resident review	The resident confirms both and keeps the materials separate if possible.
Helper handover	The helper checks the material, records scale readings and sends the receipt.

The app does not automatically treat the whole load as organic.

Where the collected waste goes

The pilot needs agreed handovers with local processing partners.

Organic waste	Suitable clean organics go to an approved composting or biogas partner. The processor confirms what it accepts.
Dry materials	Paper, plastic, glass and metal go to suitable recyclers after material checks.
Special waste	Batteries, hazardous items and unsuitable loads need a separate specialist arrangement.

Recovery income and programme benefits require actual acceptance and payment records.

Plans for different collection needs

Current pilot prices. One-month passes have no automatic renewal.

	Basic	Gold	Platinum
Price	₹10 / pickup	₹399 / month	₹469 / month
Included pickups	1	14	24
Per pickup limit	2 kg	8 kg	10 kg
Monthly allowance	Pay per pickup	112 kg	240 kg
Segregation	₹5/kg if mixed	Included	Included
Resident recovery share	0%	10% of net	15% of net

Metal counts toward limits. Specialist or hazardous waste needs separate handling.

Clear charges before and after weighing

Residents never have to guess the weight.

Standard hours	7 am to before 2 pm. No collection from 2 pm to 5 pm.
Evening hours	5 pm to 7 pm. A ₹5–₹7 surcharge depends on time, weight and sorting.
Mixed Basic waste	₹5 per measured kg for segregation. Gold and Platinum include segregation.

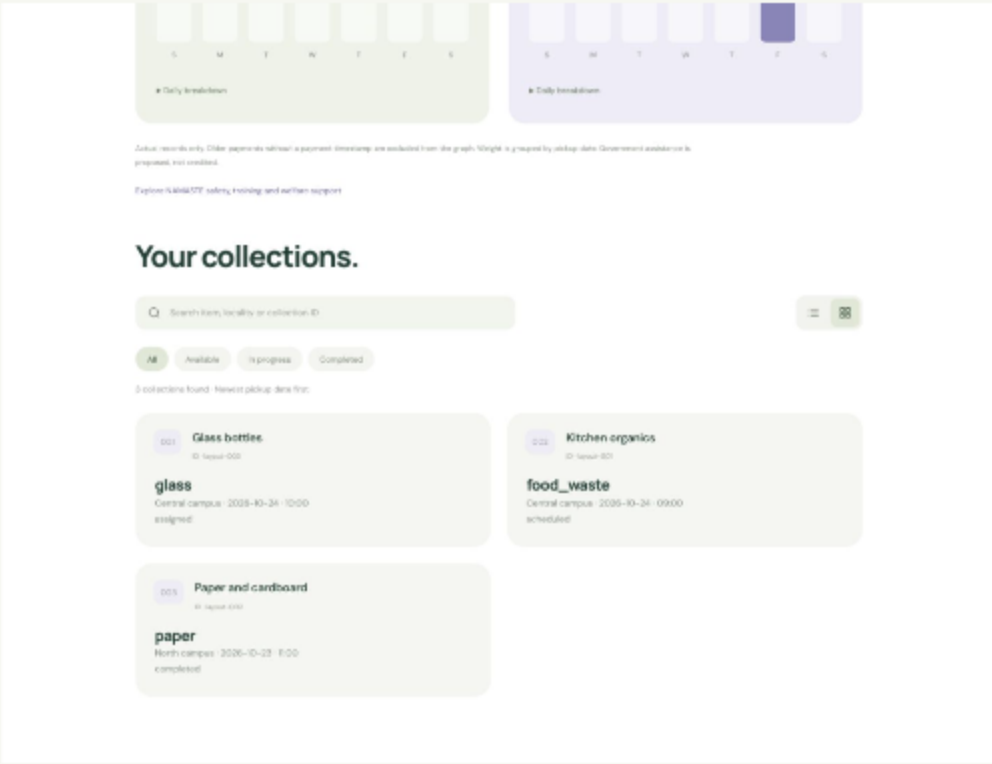
Example: 2 kg of mixed Basic waste at 10 am = ₹10 pickup + ₹10 sorting = ₹20.

Helpers get a dedicated workspace

English and Hindi support the collection workflow.

Collections have their own page

Search by item, locality or ID.
Switch between grid and list.
Accept jobs and record weight.
Track receipts and status.



Synthetic layout example

Helper earnings are visible

The pilot separates earned pay, recorded payments and outstanding amounts.

₹8

per completed pickup

+ 50% of sorting service value

+ 40% of net recovery

Earnings require a completed, accepted receipt. Operations records actual payments.

Operations knows what to do next

A separate workspace for Collections, Payments, Helpers and Issues.

A collection queue

Search records and open one collection at a time.

Clear payment steps

See earned, paid and outstanding helper amounts. Record a completed transfer.

Separate sale records

Record actual recycler income and costs, then the resident’s recovery share.

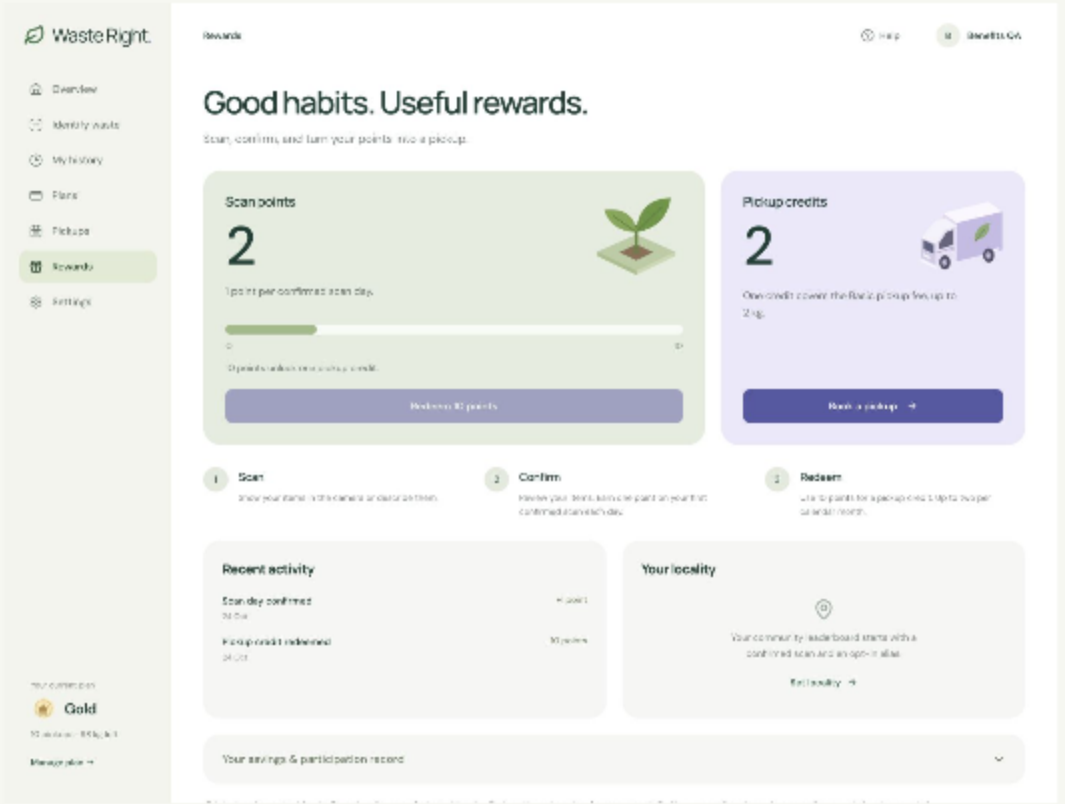
Rewards make the habit easier to keep

Earn one point on the first confirmed scan day.

10 points

unlock one pickup credit

Up to two credits per calendar month.
Each covers the Basic fee up to 2 kg.



A useful record of participation

A completed organic pickup unlocks a downloadable Waste Right certificate.

Name and participation ID

Completed organic pickups
Helper-measured organic kg
Issue date and activity snapshot

Independent issuer: Waste Right



Illustrative certificate

How Waste Right earns revenue

Customer payments and verified material recovery fund the service.

Pickup fees	Basic collections generate a fee for each completed booking.
Monthly passes	Gold and Platinum generate revenue when residents buy or renew their pass.
Material recovery	Accepted recycler or processor proceeds contribute after direct costs and shares.

Government support adds income only after eligibility and actual receipt.

A ₹10 pickup leaves a narrow margin

Illustrative base-fee calculation before travel and other costs.

₹10

customer pickup fee

₹8

base helper pay

₹2

left before other costs

Dense routes, subscriptions and actual recovery must cover transport, sorting, payment fees and overhead.

GOBARdhan: the organic-processing route

A proposed role for Waste Right: collection and feedstock aggregation.

Programme	The integrated 2026–36 CBG framework supports production and market access.
Our proposed role	Deliver suitable organic waste to a contracted biogas or CBG processing partner.
What is still required	Partner acceptance terms, project eligibility and the relevant approvals. No Waste Right award is documented.

Urban partnerships and startup funding

Two distinct routes: municipal service agreements and competitive innovation support.

SBM–Urban 2.0

Supports segregation, door-to-door collection and scientific processing.

Seek an urban local body pilot. Confirm current local approvals and procurement routes.

Swachhata Startup Challenge

The 2022 competition offered ₹25 lakh each to ten selected projects.

That window is closed. No current call or Waste Right award is verified.

A PPP requires a signed public-authority agreement; scheme alignment alone is insufficient.

Fertiliser support and the resident benefit

Support is recorded only after eligibility and actual receipt.

Published MDA terms

₹1,500 per tonne of eligible fertiliser output from BG/CBG plants.

The March 2026 update said the scheme was concluding that year. Continuation needs confirmation.

Waste Right credit policy

50% of attributable support actually received is allocated to the resident as a recorded credit.

This is our policy, not a government-set entitlement.

Raw household waste weight cannot directly determine an MDA payment.

Support for collection partners

Safe work and access to welfare matter alongside pay.

Our service responsibilities

Clear pay records and dispute handling.
Review route time and travel costs.
Plan safety equipment and training.

Potential NAMASTE support

Validated waste pickers may qualify for PPE, safety training and health coverage under scheme rules.

Waste Right signup alone does not establish government eligibility.

AI assists the collection decision

The current app combines image understanding, OCR and resident review.

Image understanding	Suggests likely materials from camera frames or a description.
OCR	Reads visible packaging text to add context.
Human confirmation	The resident confirms the list. The helper establishes actual material and weight.

Our own organic-waste model

A trained research model separates organic regions from visible contaminants.

450

labelled source images

67

held-out test images

66.13%

foreground mean IoU

IoU measures overlap between predicted and labelled regions. It is not item-identification accuracy.

One household and one fixed camera. Phone-camera validation is still needed.

A working app with a separate model service

Simple components keep the demo understandable and the model replaceable.

Web app	Next.js and React with role-based resident, helper and admin workspaces.
Application records	Local MongoDB stores accounts, bookings, receipts, benefits and payment records.
AI and payments	Configurable image AI and OCR. Razorpay checkout integration. Separate local Python inference API.

What works today

A tested prototype, with clear limits before launch.

Built in the prototype

Camera and description flow
Automatic fees and scheduling
Helper collections and earnings
Receipts, disputes and PDF certificates

Before a public launch

Live payment and bank checks
Field tests on phones and routes
Approved processor agreements
Government eligibility and identity onboarding

31 tests passed, plus isolated API integration checks and a production build.

A focused first pilot

Proposed scope: one campus or housing cluster, with one processing partner.

Start small	Target 50 households and two approved helpers for a four-week pilot.
Measure the work	Track completed pickups, route minutes, kg accepted, repeat bookings and disputes.
Measure the money	Track helper hourly take-home and contribution after every direct cost.

Pilot gate: reliable collections, positive route contribution and acceptable helper earnings.

What we need next

A pilot location and a processing partner to test the complete service.

A campus or housing association

A local waste processor and collection partners

Support to measure real costs and improve the service

Appendix: rewards and service rules

Useful details for reading the deck without a presenter.

Points and credits

One point per first confirmed scan day in India. Ten points per credit. Maximum two credits per calendar month.

Savings and history

Savings separate fee reductions, included segregation and recorded government credits. History stores scans, receipts and disputes.

Certificate and identity

Completed measured organic collections unlock the PDF. Waste Right issues it using the account name. DigiLocker and Aadhaar linking remain unconfigured.

Appendix: pricing and earnings examples

Illustrative calculations under the current pilot policy.

2 kg mixed Basic pickup	Customer pays ₹20 in standard hours. Helper earns ₹8 + ₹5 sorting share = ₹13 before any recovery share.
₹100 positive recovery pool	Helper recovery share is ₹40. Gold resident share is ₹10, or Platinum ₹15. Remaining ₹50 or ₹45 is before other business expenses.
₹100 attributable support received	Resident credit policy allocates ₹50 after completed processing, receipt and credit records. No projected grant appears as savings.

Sources and evidence

The pitch uses project records and official scheme information.

Product and business rules

Current app catalog and pricing logic
Helper pay and benefit accounting
Certificate eligibility and edge-case audit

Model evidence

Household-v2 results and model card
TU Wien dataset, CC BY 4.0
doi.org/10.48436/27k90-dvw73

Government sources: PIB and the implementing departments. GOBARdhan, SBM–Urban 2.0, MDA, NAMASTE and the 2022 startup challenge.